

Total No. of Questions : 8]

SEAT No. :

P2989

[Total No. of Pages : 2

[5669]-581

T.E. (Computer Engineering) (Semester - I)

THEORY OF COMPUTATION

(2015 Pattern)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume suitable data, if necessary.

Q1) a) Define the following terms with example.

[6]

- i) Alphabet
- ii) String
- iii) Language

b) What are the different types of Normal Forms of Context Free Grammar? Explain it with example. [6]

c) Construct a Finite Automata for the following Regular Expressions. [8]

- i) $(1+0)^*(00+11+10)$
- ii) $1(01+10)^*+0(11+10)^*$

OR

Q2) a) Simplify the following Context Free Grammar (CFG).

[6]

$S \rightarrow ASB \mid \epsilon$

$A \rightarrow aAS \mid \epsilon$

$B \rightarrow Sbs \mid A \mid bb$

$G = \{(S,A,B), (a,b), S,P : \text{Productions are given above}\}.$

b) Construct a Mealy Machine for 2's complement of a binary number. [6]

c) Construct a Deterministic Finite Automata (DFA) for the following. [8]

- i) $L = \{0,1 \mid \text{Accept all the strings ending in 00 or 11}\}.$
- ii) Accept a binary number divisible by 3.

P.T.O.

- Q3)** a) Write a short note on Halting Problem of Turing Machine. [4]
 b) Construct a Turing Machine to find 2's complement of a binary number. [6]
 c) Construct a Turing Machine to add two unary numbers. [8]

OR

- Q4)** a) Explain the power of Turing Machine over Finite Automata. [4]
 b) Construct a Turing Machine for the language $L = \{a^n b c^n \mid n \geq 1\}$ [6]
 c) Construct a Turing Machine which accepts odd length palindrome over the $\Sigma = \{a, b\}$. [8]

- Q5)** a) Define Pushdown Automata (PDA)? What are the different types of PDA? Explain any two applications of PDA? [6]
 b) Design a PDA for accepting a language $L = \{a^n b^m c^n \mid m, n \geq 1\}$ [6]
 c) Explain the acceptance of language by PDA - [4]
 i) By Final State
 ii) By Empty Stack

OR

- Q6)** a) Design a PDA for accepting language $L = \{W c W^R \mid W \in (a,b)^*\}$. [6]
 b) Give Context Free Grammar (CFG) generating the language accepted by the PDA $M = \{(q_0, q_1), (a, b), \delta, q_0, Z_0, q_1\}$ where δ is as follows.
 $\delta(q_0, a, Z_0) \rightarrow (q_0, XZ_0)$
 $\delta(q_0, a, X) \rightarrow (q_0, XX)$
 $\delta(q_0, b, X) \rightarrow (q_1, \epsilon)$
 $\delta(q_1, b, X) \rightarrow (q_1, \epsilon)$
 $\delta(q_1, \epsilon, Z_0) \rightarrow (q_1, \epsilon)$
 c) Prove that "Let L be a language accepted by deterministic PDA, then the complement of L, can also be accepted by, deterministic PDA. [4]

- Q7)** a) Write a short note on Post Correspondence Problem. [6]
 b) What are Tractable and Intractable problems? Explain it. [6]
 c) Differentiate between P-class problems and NP-class problems. [4]

OR

- Q8)** a) What is Travelling Salesman Problem? Justify that it is a NP-class Problem. [6]
 b) Write a short note on Node-Cover Problem. [6]
 c) What is Polynomial Time Reduction? Explain It with suitable example. [4]

